

Individual reflections on the project HR Analytics Proof of Concept

Technical steps

First off, we created a Snowflake warehouse and database named group4_wh and group4_db, the three different schemas Staging, warehouse and mart and different roles (DLT/DBT/Analyzing) to conduct operations in the warehouse.

Thereafter, we downloaded data-loading-tool (DLT) and used this in a python script named load_jo_ads.py to load data from <https://jobsearch.api.jobtechdev.se> into the staging layer of our database, selecting by three different occupational fields.

In the next step we used data-build-tool (DBT) to enable us to create a star schema within our database. Where we first create a source layer (SRC) of relevant data in five different tables. We thereafter use some of this data in a created fact table (fct_jobs_ads) where we link the SRC tables to the fact table by creating primary surrogate keys. In the next step, for each SRC table we create a dimensional table (DIM) where we use the same surrogate key to enable linking between the fact table and dimensional tables.

In the final mart layers we use data columns from the fact and dim tables ready to be used for our dashboard by creating a different mart layer for each occupational field for simplicity.

The SRC enabled us to have a “staging layer” where the data is untouched so we can trace back to if we would need to if data would change in the fact or mart layers for any reason. The tables of dim layers becomes reliable sources of data with its surrogate keys and the fact table links to all different dimensions – making joins operations easier. The final mart layer is easy to use for the dashboard with prepared relevant columns.

In the next step we load our data into our Streamlit dashboard (dashboard.py) with snowflake connector in the script connect_data_warehouse.py, and start the dashboard itself by running the script run_dashboard.py which starts the dashboard.

Reflection on the team work and dashboard

Initially we worked by distance but tried to have daily standups. I general all participants tried to contribute to every stage of the project for learning purposes, because if you divide different parts of the project between us – you would not get a full understanding of the project.

With that said, this was not always possible, and as an example I by myself created the warehouse, database and different roles in Snowflake and made the DLT process – but informed Alex and Eyoub on the process and educated them about the script so they got

an understanding. And as another example – they made a bigger effort in creating the dashboard meanwhile I worked with DBT testing and Documentation in terms of efficiency, but where we educated each other on what we had done individually.

In the last 7-10 of the project we worked in the school together more or less full days where we worked a lot with mob programming, where we circulated positions for who was the designated driver and navigator. We found this to be a positive way for working, especially in more complicated parts of the project e.g. creating the different dimensions and surrogate keys as we discussed and solved different problems better together through discussion.

I liked to work with Alex and Eyoub but if we would work with a future project I think it would be good for us to start a bit earlier and have 1-2 extra days where we could fine tune our dashboard more and at least add more relevant KPI ´s. E.g. KPI ´s such as last day for application, incoming ad ´s set over time to see a timeline and map for a better overview like some of our classmates had.